Unit 4 · Outcome 2 · Energy Sources

The Two Laws That Decide Every Energy Choice

KK06 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.

WHY A PHYSICS RULE BELONGS IN ENVIRONMENTAL SCIENCE

01 Why a physics rule belongs in Environmental Science

The study design asks you to explain the implications of the first and second laws of thermodynamics when making energy choices. Not the maths of efficiency (that was KK05) — the consequences. These two laws are the reason a coal plant throws away two-thirds of its fuel, the reason "free energy" machines are scams, and the reason cutting waste beats building more supply.

> ◆ **NOTE**
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> The Tarra Cove decision. The town of Tarra Cove, on the edge of Bass Strait, has to power its new desalination plant. The council is handed three options: pipe in electricity from the inland Wirndook brown-coal station, build local solar + battery, or run a gas peaker. Every councillor wants the "most efficient" choice — but none of them can escape the two laws. By the end of this module, you'll be the one explaining why.

THE TWO LAWS, STATED CLEANLY

02 The two laws, stated cleanly

Learn both the formal statement (for the high-mark questions) and the plain-English version (so you actually understand it).

"Energy cannot be created or destroyed, only transformed from one form into another. The total energy of an isolated system is constant."

Whatever energy you start with, you still have — it has just changed form. If only 33 units come out useful, the other 67 didn't vanish; they became low-grade heat spread into the surroundings.

"In any energy conversion, the total entropy (disorder) of the system and surroundings increases. Some energy is always degraded to low-grade, dispersed heat that can no longer do useful work."

You can account for all the energy (first law), but you can never use all of it. A bit becomes useless waste heat every single step, and it only flows one way: hot → cold, ordered → disordered.

> ◆ **NOTE**
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> The one-line memory hook: First law = you can't win (you'll never get out more than you put in). Second law = you can't break even (you'll always get out less than you put in). Together: you can't get out of the game — there's no perfect machine.

FIRST LAW: THE ENERGY IS STILL THERE CONSERVATION

03 First law: the energy is still there

The most common misread is thinking inefficiency means energy "lost". It isn't lost — it's relocated and reshaped. When the Wirndook turbine is 40% efficient, the missing 60% is now warm cooling water in the Kalduke River, warm flue gas up the stack, and friction heat in the bearings. Add it all up and you're back to 100%.

◈ **ANALOGY**

Analogy · the energy bank account Think of energy like money in a sealed economy. You can move it between accounts (chemical → heat → motion → electricity) but you can never print new notes or burn them out of existence. Every cent is always somewhere. An "energy audit" of any machine must balance to exactly 100% — if it doesn't, you've missed a stream of waste heat.

The implication for choices: there is no such thing as a free lunch. Any device claiming to output more energy than it consumes — an "over-unity" or "perpetual motion" generator — breaks the first law and cannot exist. When an energy pitch sounds too good, the first law is your fraud detector.

◆ **NOTE**

Where does the energy "go" in your phone charger? A charger pulling 10 W but delivering 8 W to the battery isn't destroying 2 W — that's why the plug feels warm. The first law says those 2 W must appear somewhere, and the second law says "somewhere" is almost always low-grade heat.

SECOND LAW: ENERGY QUALITY ONLY EVER FALLS ENTROPY ↑

04 Second law: energy quality only ever falls

The first law treats every joule as equal. The second law says they're not. High-grade energy is concentrated and ordered — it can do work (electricity, the chemical energy in petrol, a tank of compressed steam). Low-grade energy is dispersed and lukewarm — heat spread thinly through the air or ocean. Every conversion shifts a slice of high-grade into low-grade, and the universe's total entropy (disorder) rises.

Crucially, this is a one-way street. Heat flows from hot to cold on its own, but never the reverse without you spending more high-grade energy (that's what a fridge or heat pump does — it pumps heat uphill, but only by consuming electricity). You cannot un-disperse waste heat back into a neat tank of useful energy for free.

Start: energy is tightly packed and hot in one corner — it can do work. Press Let it run to watch the second law happen.

◈ **ANALOGY**

Analogy · the drop of cordial Drip red cordial into a glass of water and it spreads until the whole glass is pale pink. It never spontaneously gathers back into a single drop. Energy does the same: concentrated → spread out, and never spontaneously the reverse. Stirring it back together would cost you energy — and you'd still never get the clean drop returned. That irreversibility is the second law.

WHY NO POWER STATION IS EVER 100% EFFICIENT

05 Why no power station is ever 100% efficient

Put the two laws together and you get the headline result of this whole chapter. Any source that works by burning fuel to make heat, then converting heat to motion — coal, gas, oil, nuclear, even biomass — is a heat engine, and heat engines are hard-capped by the second law. You can only extract work from heat as it flows from a hot source to a cold sink, and a large fraction must be dumped to the cold side as waste. That waste isn't bad engineering; it's physics.

Energy pathway	Heat engine?	Typical real efficiency	Why
Brown-coal electricity (Wirndook)	Yes	~25–33%	Wet, low-energy lignite + unavoidable heat-engine losses
Combined-cycle gas	Yes	~50–60%	Captures more of the heat in two stages, but still capped
Solar PV	No	~15–22%	Not a heat engine, but photons missed/lost as heat
Hydro / wind	No	~80–90%	Converts existing motion directly — very few losses
Solar hot water (direct)	No	~60–70%	Sunlight → heat → hot water, almost no conversion steps

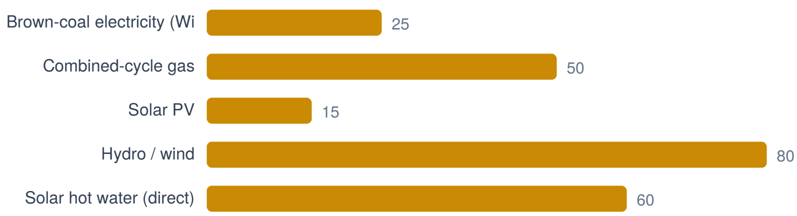


Figure · Energy pathway compared at a glance.

◆ **NOTE**

Read the table like an examiner. Notice the heat engines cluster low and the direct converters cluster high. That isn't a coincidence — it's the second law sorting them. A strong exam answer names both laws: the first explains where the missing energy went (waste heat), the second explains why it's unavoidable (entropy must rise).

ENERGY QUALITY: MATCH THE TOOL TO THE JOB

06 Energy quality: match the tool to the job

Because the second law makes high-grade energy precious, a smart energy choice matches the quality of the energy to the quality the job actually needs. Using premium electricity to do a job that only needs low-grade warmth is like using filtered drinking water to flush a toilet — it works, but you've wasted something valuable.

Electricity, chemical fuels, compressed gas. Can be converted into almost any form. The most useful and most "expensive" in second-law terms — every conversion that made it already paid a tax.

Waste heat from a power station, lukewarm air, the warmth in the ocean. Plenty of it exists, but it's spread too thin to drive a turbine or a motor. Mostly destined to be lost.

The clever move is cogeneration (also called combined heat and power): instead of fighting the waste heat from a power station, you capture it and pipe it to heat nearby buildings or run an industrial process. You can't beat the second law — but you can stop throwing the loser's share away.

THE IMPLICATIONS — MAKING THE TARRA COVE CHOICE

07 The implications — making the Tarra Cove choice

This is the exact skill the study design names: turning the two laws into decisions. Every conversion step in a chain pays a second-law tax, and the taxes multiply. So one rule dominates: fewer conversions, and use the source closest to the form you need.

The five practical implications you should be able to argue in an exam:

- Each step sheds waste heat (2nd law), so chains with fewer links keep more useful energy. Direct solar hot water beats coal-fired resistive heating for the same warmth.
- Coal, gas, oil and nuclear are capped low by the second law. If a choice is dominated by heat engines, large unavoidable losses are locked in.
- Don't burn high-grade electricity for low-grade warmth. A heat pump moving existing heat beats a resistive heater making new heat.
- You can't stop the loss, but you can reuse it for heating instead of dumping it to a river or the sky.
- Every joule not used avoids all the upstream conversion losses — so efficiency and reduced demand do more than new supply.
- "Over-unity" and perpetual-motion claims break the first law — no investigation needed, the physics already says no.

WHICH LAW IS IT? — SORT THE STATEMENTS

08 Which law is it? — sort the statements

◇ **ACTIVE RECALL**

The single most common exam slip is naming the wrong law. Tap a statement, then tap the bin it belongs in. First = about energy being conserved / where it went. Second = about quality, waste heat, entropy or direction. Both = needs the two laws together.

DECODE THE COMMAND WORD

09 Decode the command word

◈ **COMMAND WORDS**

The verb in the question tells you how much to write. Tap each to see what the marker wants for a thermodynamics question.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

SIX TRAPS THAT COST REAL MARKS

10 Six traps that cost real marks

⚠ **EXAM TRAPS**

Each of these has burned students in real VCAA-style answers. Open them and steal the fix.

P·E·L WRITING TRAINER

11 P·E·L writing trainer

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✎ **P·E·L WRITING**

P oint → E xplain → L ink. Write a 3-mark answer below. As you type, the targets light up when you hit the ideas a marker is hunting for. This is live keyword detection — it rewards substance, not length.

SAME QUESTION, TWO ANSWERS

12 Same question, two answers

A weak answer and a strong answer to the same 4-mark prompt. The highlights show exactly where marks are won and lost.

Question. The Tarra Cove council is choosing between coal-fired electricity (then used in resistive heaters) and a direct solar hot-water system to warm water for its swimming pool. Using the laws of thermodynamics, justify which is the more efficient choice. (4 marks)

● **NOTE**

Names no law. "Renewable / doesn't pollute" is a different dotpoint — it doesn't answer a thermodynamics question. "Wastes energy" is asserted, never explained. No mention of conversion steps or energy quality.

● **NOTE**

Names both laws and uses them correctly, tracks the conversion chain, invokes energy quality matching, and lands a clear justified decision. Full marks.

PRACTICE EXAM QUESTIONS / 20 MARKS

13 Practice exam questions

★ **PRACTICE & SELF-MARK**

Five VCAA-style questions, scaling from 2 to 6 marks. Write your answer first, reveal the model answer and marking guide, then award yourself stars. Your scores save automatically and feed the dock below.

ONE-SCREEN CHEAT SHEET

14 One-screen cheat sheet

- Energy is conserved — created/destroyed by nothing, only transformed.
- Inefficiency ≠ loss; the energy became low-grade waste heat in the surroundings.
- Implication: no perpetual motion / over-unity machine can exist.

- Every conversion increases entropy; some energy degrades to dispersed heat.
- Energy quality always falls; heat flows hot→cold, never the reverse for free.
- Implication: $\eta < 100\%$ always; heat engines are capped low.

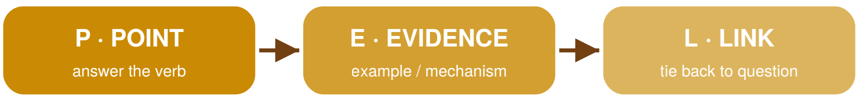
- Fewer conversion steps = more useful energy kept.
- Match quality to task — don't waste electricity on warmth.
- Cogeneration reuses waste heat; saving energy skips all upstream losses.

- First = where did it go; Second = why can't I reuse it.
- "Renewable / clean" is a different dotpoint — don't answer with it here.
- Always name the law before you use it.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Solar is better because it's renewable and doesn't pollute. Coal is bad for the environment and wastes a lot of energy. The solar option is more efficient and cheaper so the council should pick it. Names no law. "Renewable / doesn't pollute" is a different dotpoint — it doesn't answer a thermodynamics question. "Wastes energy" is asserted, never explained. No mention of conversion steps or energy quality.

✓ STRONG / MODEL ANSWER

Heating water needs only low-grade thermal energy. The coal path makes electricity through many conversions (chemical → heat → motion → electricity → heat), and by the second law each step degrades some energy to waste heat, so only ~33% of the coal's energy survives. The first law confirms the rest isn't destroyed — it's lost as heat to the surroundings. The solar system converts sunlight directly into heat in far fewer steps, so much less energy is degraded. Because it matches a low-grade need with a low-grade source, the solar option is the more efficient choice. Names both laws and uses them correctly, tracks the conversion chain, invokes energy quality matching,

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.